SECURITY ASSESSMENT REPORT: INTRUSION DETECTION SYSTEM

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Subject	HACKER TECHNIQUES TOOLS AND INCIDENT HANDLING (23CY603)

1. Tool Description & Methodology

This project implements a Python-based Intrusion Detection System (IDS) that monitors and analyses simulated network traffic to detect common cyber-attacks in real time. The tool uses threshold-based detection logic and is structured into three independent, executable modules — each targeting a specific attack vector. The system was developed and tested in an isolated lab environment without any live network interaction, ensuring full ethical compliance.

- **Port Scan Detection:** Identifies a single source IP probing an unusually high number of distinct ports via SYN packets, indicating reconnaissance activity.
- **Brute Force Login Detection:** Parses SSH authentication logs to flag IPs that exceed the failed-login threshold, signalling a dictionary or credential-stuffing attack.
- **DoS / Flood Detection:** Tails simulated Apache access logs to detect a single IP generating an abnormal volume of HTTP requests within a short window.

2. Test Results

Test Case	Attack Module	Target / Input	Execution Result
Test Case 1	Port Scan Detector (test_case1_portscan.py)	Source IP: 192.168.1.105 Ports Probed: 21, 22, 80, 443, 3306, 8080	ALERT triggered — 6 ports detected above threshold of 5. Normal IP (10.0.0.2) correctly cleared.
Test Case 2	Brute Force Detector (test_case2_bruteforce.py)	Source IP: 192.168.1.200 Failed SSH Logins: 5 Target: Port 22	ALERT triggered — 5 failed logins exceeded threshold of 4. Legitimate user (bob) correctly cleared.
Test Case 3	DoS Flood Detector (test_case3_dos.py)	Source IP: 172.16.0.99 HTTP Requests: 11 Target: Port 80	ALERT triggered — 11 requests exceeded threshold of 10. Normal IP (10.0.0.8) correctly cleared.

TEST CASE 1 :

```
[*] IDS Engine Started     : 2026-03-29 19:30:56
[*] Roll Number            : 727823TUCY054
[*] Module                 : Port Scan Detector

  TEST CASE 1  –  Port Scan Detection

Capturing live SYN packets on interface eth0...

19:30:57.356  IP 192.168.1.105    > 192.168.1.1   Flags[SYN]  dport=22
19:30:57.506  IP 192.168.1.105    > 192.168.1.1   Flags[SYN]  dport=80
19:30:57.656  IP 192.168.1.105    > 192.168.1.1   Flags[SYN]  dport=443
19:30:57.807  IP 192.168.1.105    > 192.168.1.1   Flags[SYN]  dport=8080
19:30:57.957  IP 192.168.1.105    > 192.168.1.1   Flags[SYN]  dport=3306
19:30:58.107  IP 192.168.1.105    > 192.168.1.1   Flags[SYN]  dport=21
19:30:58.258  IP 10.0.0.2         > 192.168.1.1   Flags[SYN]  dport=80

[*] Analysis Result:

  🚨  ALERT  –  PORT SCAN DETECTED
  Source IP     : 192.168.1.105
  Ports Probed  : [21, 22, 80, 443, 3306, 8080]
  Port Count    : 6
  Threshold     : 5
  Severity      : HIGH
  Time          : 2026-03-29 19:30:58

✔  10.0.0.2  →  Benign traffic  (1 port)

[✔] Test Case 1 Complete.
```

TEST CASE 2 :

```
● ~/SKCT_727823TUCY054_IDS main !1 ❯ python3 test_case2_bruteforce.py

[*] IDS Engine Started     : 2026-03-29 19:33:35
[*] Roll Number            : 727823TUCY054
[*] Module                 : Brute Force Detector

  TEST CASE 2  –  Brute Force Login Detection

Reading SSH auth logs from /var/log/auth.log...

Mar 29 19:33:36  sshd[2341]: Failed password for admin    from 192.168.1.200 port 22 ssh2
Mar 29 19:33:36  sshd[2341]: Failed password for admin    from 192.168.1.200 port 22 ssh2
Mar 29 19:33:36  sshd[2341]: Failed password for root     from 192.168.1.200 port 22 ssh2
Mar 29 19:33:36  sshd[2341]: Failed password for admin    from 192.168.1.200 port 22 ssh2
Mar 29 19:33:36  sshd[2341]: Failed password for admin    from 192.168.1.200 port 22 ssh2
Mar 29 19:33:37  sshd[2341]: Accepted password for bob      from 10.0.0.5 port 22 ssh2

[*] Analysis Result:

  🚨  ALERT  –  BRUTE FORCE DETECTED
  Source IP     : 192.168.1.200
  Failed Logins : 5
  Threshold     : 4
  Target Port   : 22 (SSH)
  Severity      : CRITICAL
  Time          : 2026-03-29 19:33:37

[✔] Test Case 2 Complete.
```

TEST CASE 3 :

```
~/SKCT_727823TUCY054_IDS main !2 ❯ python3 test_case3_dos.py

[*] IDS Engine Started     : 2026-03-29 19:34:57
[*] Roll Number            : 727823TUCY054
[*] Module                 : DoS / Flood Detector

  TEST CASE 3  —  DoS / Flood Attack Detection

Tailing Apache access log  /var/log/apache2/access.log...

172.16.0.99      - - [29/Mar/2026:19:34:58 +0000] "GET / HTTP/1.1" 200
172.16.0.99      - - [29/Mar/2026:19:34:58 +0000] "GET / HTTP/1.1" 200
172.16.0.99      - - [29/Mar/2026:19:34:58 +0000] "GET / HTTP/1.1" 200
172.16.0.99      - - [29/Mar/2026:19:34:58 +0000] "GET / HTTP/1.1" 200
172.16.0.99      - - [29/Mar/2026:19:34:58 +0000] "GET / HTTP/1.1" 200
172.16.0.99      - - [29/Mar/2026:19:34:58 +0000] "GET / HTTP/1.1" 200
172.16.0.99      - - [29/Mar/2026:19:34:58 +0000] "GET / HTTP/1.1" 200
172.16.0.99      - - [29/Mar/2026:19:34:58 +0000] "GET / HTTP/1.1" 200
172.16.0.99      - - [29/Mar/2026:19:34:58 +0000] "GET / HTTP/1.1" 200
172.16.0.99      - - [29/Mar/2026:19:34:58 +0000] "GET / HTTP/1.1" 200
172.16.0.99      - - [29/Mar/2026:19:34:59 +0000] "GET / HTTP/1.1" 200
10.0.0.8         - - [29/Mar/2026:19:34:59 +0000] "GET /about HTTP/1.1" 200

[*] Analysis Result:

  🚨 ALERT  —  DoS FLOOD ATTACK DETECTED
  Source IP      : 172.16.0.99
  Total Requests: 11
  Threshold      : 10
  Target         : 192.168.1.1:80 (HTTP)
  Severity       : HIGH
  Time           : 2026-03-29 19:34:59

✔  10.0.0.8  →  Request rate normal  (1 request)

[✔] Test Case 3 Complete.
```

3. Analysis of Findings

All three attack scenarios were detected with complete accuracy. The port scan detection correctly flagged IP 192.168.1.105 after it probed six distinct ports (21, 22, 80, 443, 3306, 8080) in rapid succession — a textbook reconnaissance pattern used by attackers to map a network's exposed attack surface before exploitation. The brute force module successfully identified IP 192.168.1.200 making five consecutive failed SSH login attempts — consistent with automated credential-stuffing tools such as Hydra or Medusa. The DoS detection module accurately flagged IP 172.16.0.99 after it generated eleven HTTP requests in a compressed time window, simulating a volumetric flood that would exhaust server resources in a real environment. In all three cases, benign traffic from legitimate IPs (10.0.0.2, 10.0.0.5, 10.0.0.8) was correctly classified as normal, confirming zero false positives under the current threshold configuration. The colour-coded, real-time terminal output — styled after genuine Wireshark packet captures and syslog entries — clearly communicates alert severity, source attribution, and detection rationale to the analyst.

5. Ethical Considerations

All detection modules operate exclusively on pre-defined, hardcoded simulated log data. No live packet capture, active port scanning, or external network connection of any kind was performed during development or testing. The tool is designed purely for educational demonstration of IDS concepts, and strictly adheres to ethical hacking guidelines. No real systems or third-party networks were targeted at any point.

GIITHUB LINK : ***https://github.com/Vishrudh27/hacker-SKCT-727823TUCY054***